

LA HW

Included exercises: 3, 7, 11, 15, 21, 25, 29, 37, 41, 53, 55, 57, 59, 63, 67, 71

Exercises 1–16: Gaussian Elimination

Determine whether each system is consistent. If it is consistent, find its general solution.

Original Exercise 3

$$\begin{aligned}x_1 - 2x_2 &= -6, \\ -2x_1 + 3x_2 &= 7.\end{aligned}$$

Original Exercise 7

$$\begin{aligned}x_1 - 2x_2 - x_3 &= -3, \\ 2x_1 - 4x_2 + 2x_3 &= 2.\end{aligned}$$

Original Exercise 11

$$\begin{aligned}x_1 + 3x_2 + x_3 + x_4 &= -1, \\ -2x_1 - 6x_2 - x_3 &= 5, \\ x_1 + 3x_2 + 2x_3 + 3x_4 &= 2.\end{aligned}$$

Original Exercise 15

$$\begin{aligned}x_1 - x_3 - 2x_4 - 8x_5 &= -3, \\ -2x_1 + x_3 + 2x_4 + 9x_5 &= 5, \\ 3x_1 - 2x_3 - 3x_4 - 15x_5 &= -9.\end{aligned}$$

Exercises 17–26: Values Producing InconsistencyDetermine the values of r , if any, for which the given system is inconsistent.**Original Exercise 21**

$$\begin{aligned}x_1 - 3x_2 &= -2, \\ 2x_1 + rx_2 &= -4.\end{aligned}$$

Original Exercise 25

$$\begin{aligned}x_1 - x_2 + 2x_3 &= 4, \\ 3x_1 + rx_2 - x_3 &= 2.\end{aligned}$$

Exercises 27–34: Number of SolutionsFind the values of r and s yielding (a) no solution, (b) exactly one solution, or (c) infinitely many solutions.**Original Exercise 29**

$$x_1 + 2x_2 = s, \quad -4x_1 + rx_2 = 8.$$

Exercises 35–42: Rank and Nullity

Find the rank and nullity of each matrix.

Original Exercise 37

$$\begin{bmatrix} -2 & 2 & 1 & 1 & -2 \\ 1 & -1 & -1 & -3 & 3 \\ -1 & 1 & -1 & -7 & 5 \end{bmatrix}.$$

Original Exercise 41

$$\begin{bmatrix} 1 & -2 & 0 & -3 & 1 \\ 2 & -4 & -1 & -8 & 8 \\ -1 & 2 & 1 & 5 & -7 \\ 0 & 0 & 1 & 2 & -6 \end{bmatrix}.$$

Exercises 53–72: True or False

Determine whether each statement is true or false.

Original Exercise 53 A column of a matrix A is a pivot column if the corresponding column in the reduced row echelon form of A contains the leading entry of some nonzero row.

Original Exercise 55 When the forward pass of Gaussian elimination is complete, the original matrix has been transformed into one in row echelon form.

Original Exercise 57 The rank of a matrix equals the number of pivot columns in the matrix.

Original Exercise 59 There exists a 5×8 matrix with rank 3 and nullity 2.

Original Exercise 63 If the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the rank of $[A \ \mathbf{b}]$ is greater than the rank of A .

Original Exercise 67 If A is a matrix with rank k , then the vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k$ appear as columns of the reduced row echelon form of A .

Original Exercise 71 The third pivot position in a matrix lies in column 3.